appropriately named Big Rip? Not only would it be a kind of destruction from which there is no escape, quantum-mechanical fluctuation or not, it would be one that could tear apart the very fabric of reality, rendering any thinking creatures in the cosmos helpless as they watch their universe being ripped open around them.

This alarming possibility is hardly an outlandish fringe idea. In fact, the best cosmological data we have not only fails to rule it out, but, from some perspectives, slightly prefers it. So it's worth spending a bit of time exploring what, exactly, it would do to us.

A COSMOLOGICAL NONCONSTANT

Dark energy is often assumed to be a cosmological constant that stretches space out, accelerating cosmic expansion by imbuing the universe with some inherent inclination for swelling. On large scales, this is a pretty good description. But within galaxies, solar systems, or in the close vicinity of organized matter generally, a cosmological constant has no effect. It can be more properly thought of as a force for isolation—if two galaxies are already distant from one another, they get more distant, and individual galaxies, clusters, or groups of galaxies find themselves more and more alone as time goes on. They also form a bit more slowly in the presence of a cosmological constant than they otherwise would. What the cosmological constant *cannot* do is break apart anything that is already, in any sense, a coherent structure. *What therefore gravity hath joined together, let not a cosmological constant put asunder.*

The reason for this small mercy of the cosmological constant (which, to be fair, does still destroy the whole universe eventually) lies in the "constant" part of the story. If dark energy is a cosmological constant, its defining feature is that the density of dark energy in any given part of space is constant over time, even as space expands. The expansion rate isn't constant, just the density of the stuff itself, in any given volume of space. This makes sense in a way, if every bit of space is automatically assigned a set amount of dark energy within it, but it's still *super weird,* because it means that as space gets bigger, the amount of dark energy increases to keep the density constant. It also means that if you draw a sphere of a given size anywhere in the universe and measure the amount of dark energy inside the sphere, and then do the same at some future time, you'll always get the